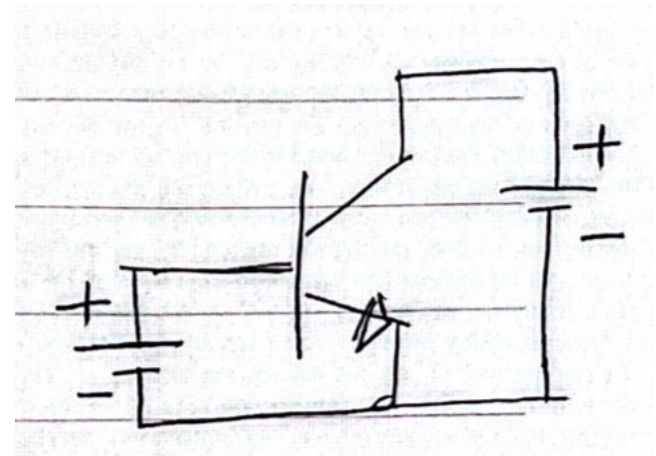
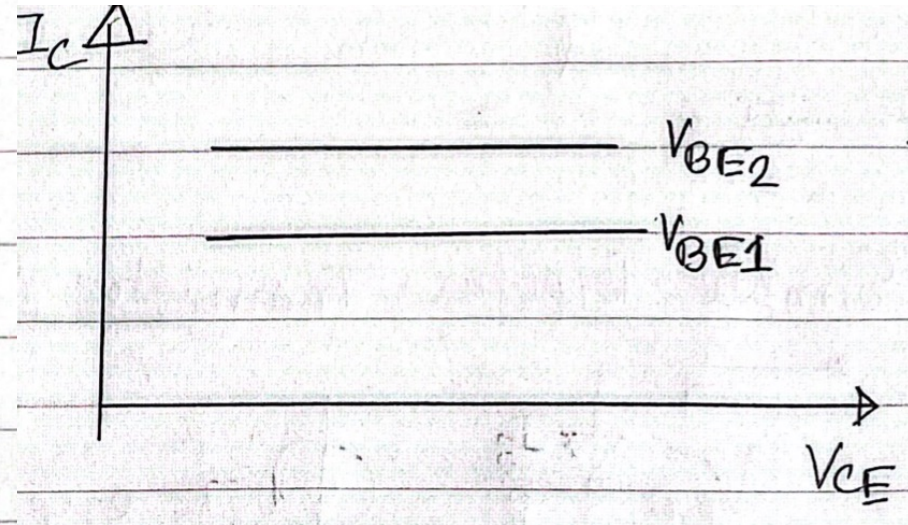
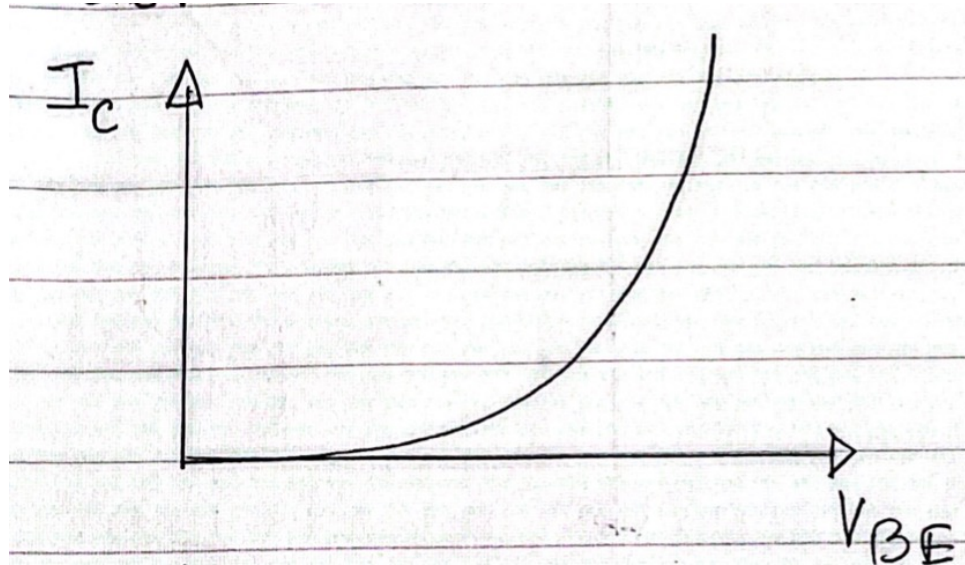
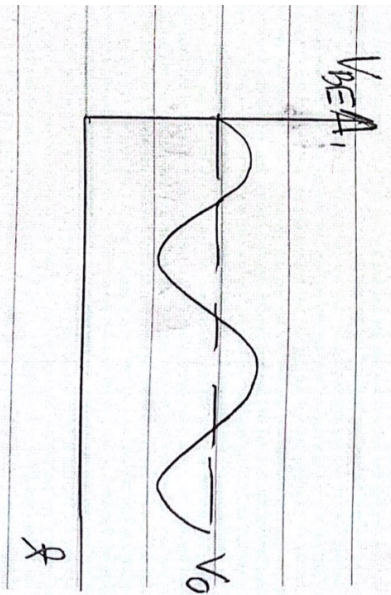
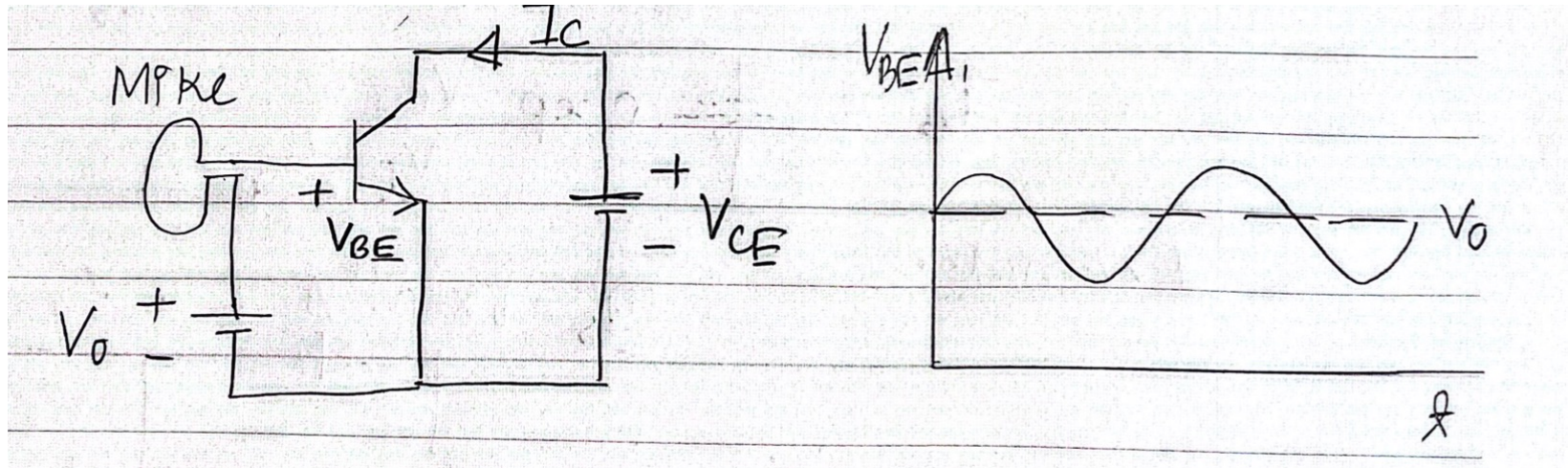


Time Response and I-V Characteristics

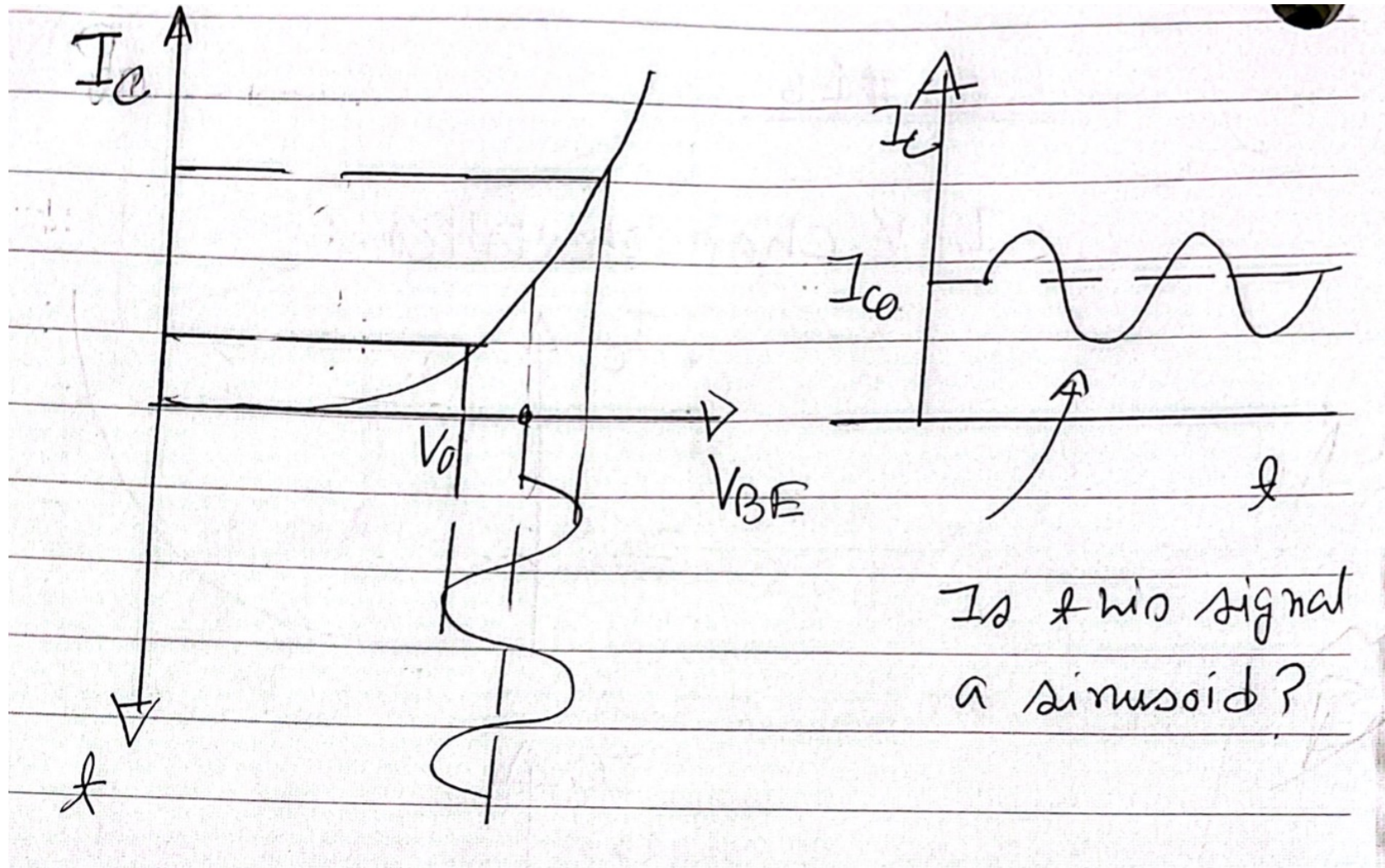
I-V Characteristics



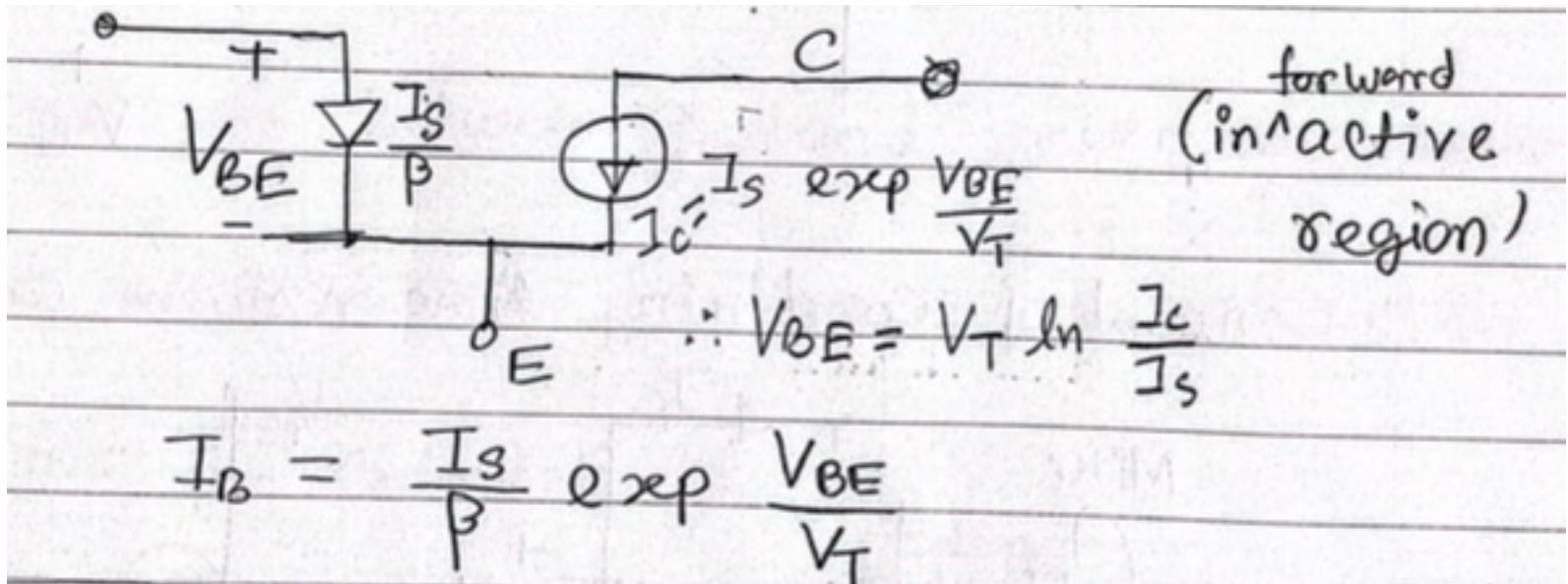
Time Response and I-V Characteristics



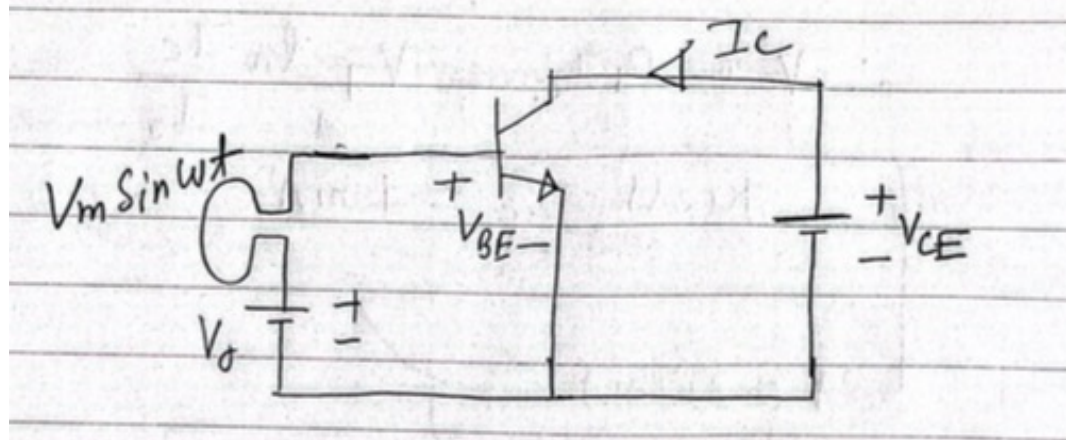
Time Response and I-V Characteristics



Bipolar Transistor Large Signal Model



Bipolar Transistor Large Signal Model

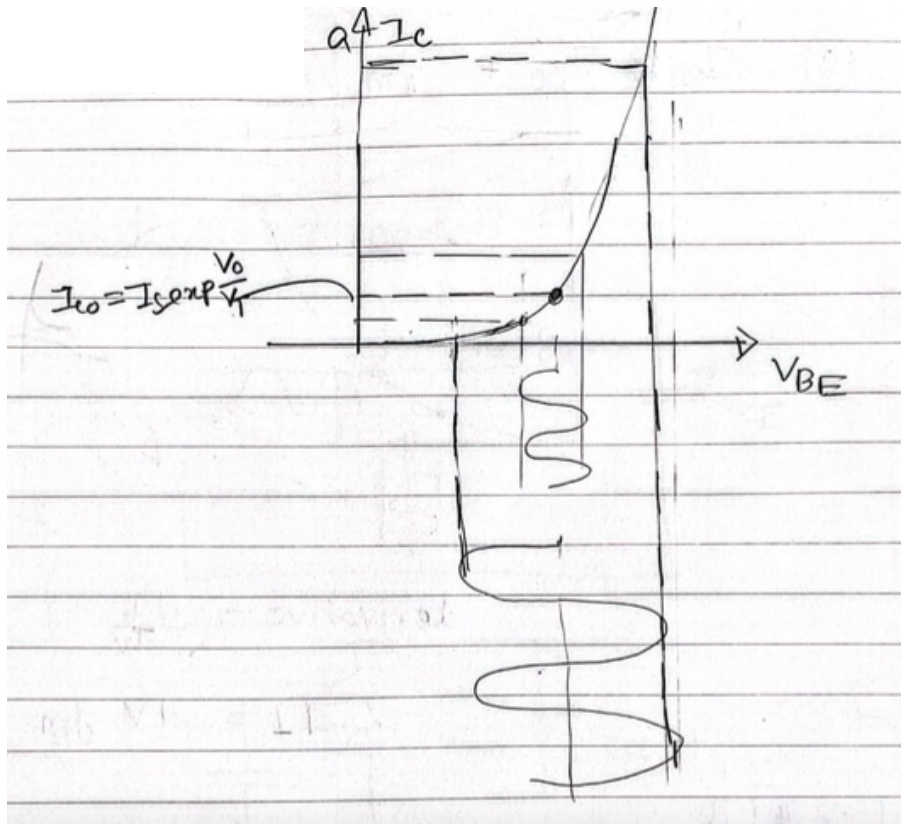


$$V_{BE} = V_0 + V_m \sin \omega t$$

$$\Rightarrow I_C = I_S \exp \frac{V_0 + V_m \sin \omega t}{V_T}$$

“Large Signal” operation: the signal is arbitrarily large.

Bipolar Transistor Small Signal Model



$$I_c = I_s \exp \frac{V_0}{V_T} \exp \frac{V_m \sin \omega t}{V_T}$$

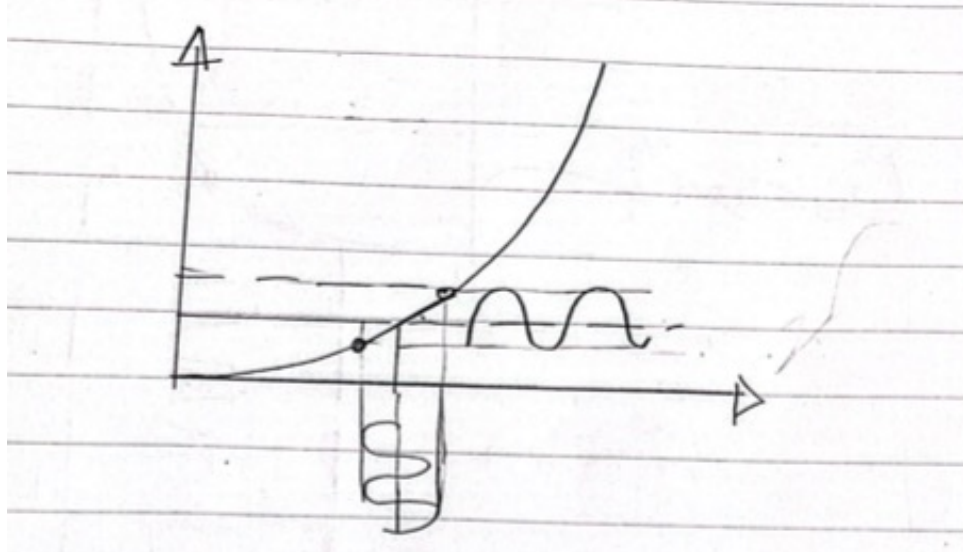
$$e^{\epsilon} \approx 1 + \epsilon \quad \text{if } \epsilon \ll 1$$

$$\begin{aligned} I_c &\approx I_s \exp \frac{V_0}{V_T} \left(1 + \frac{V_m \sin \omega t}{V_T} \right) \\ &= I_{c0} + \frac{I_{c0}}{V_T} V_m \sin \omega t \end{aligned}$$

Bipolar Transistor Small Signal Model

① I_c varies sinusoidally

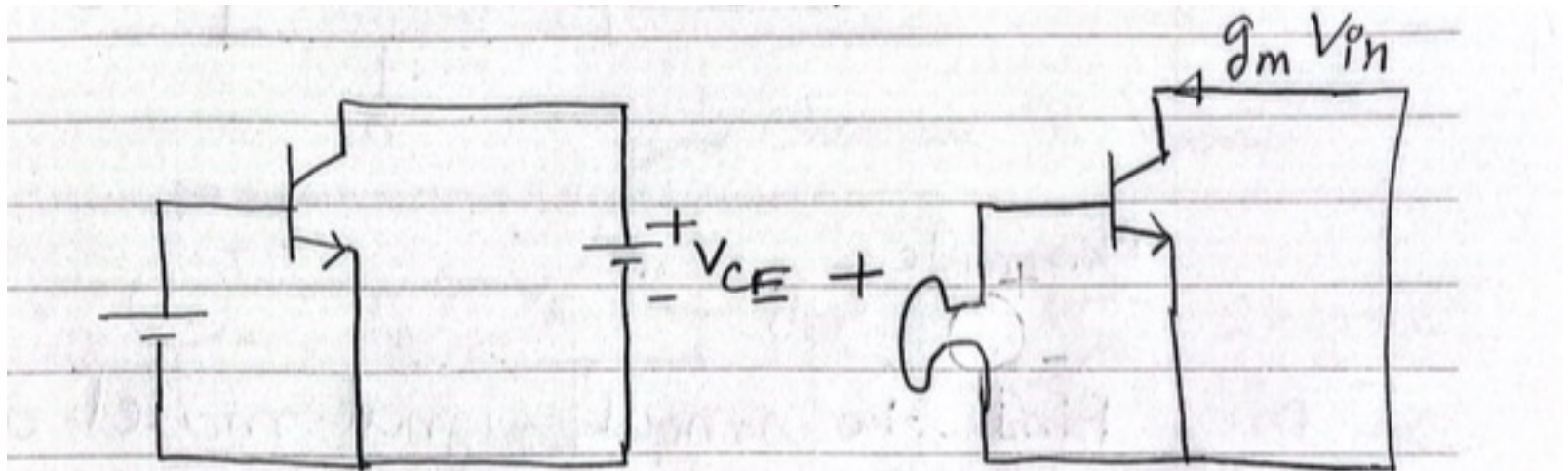
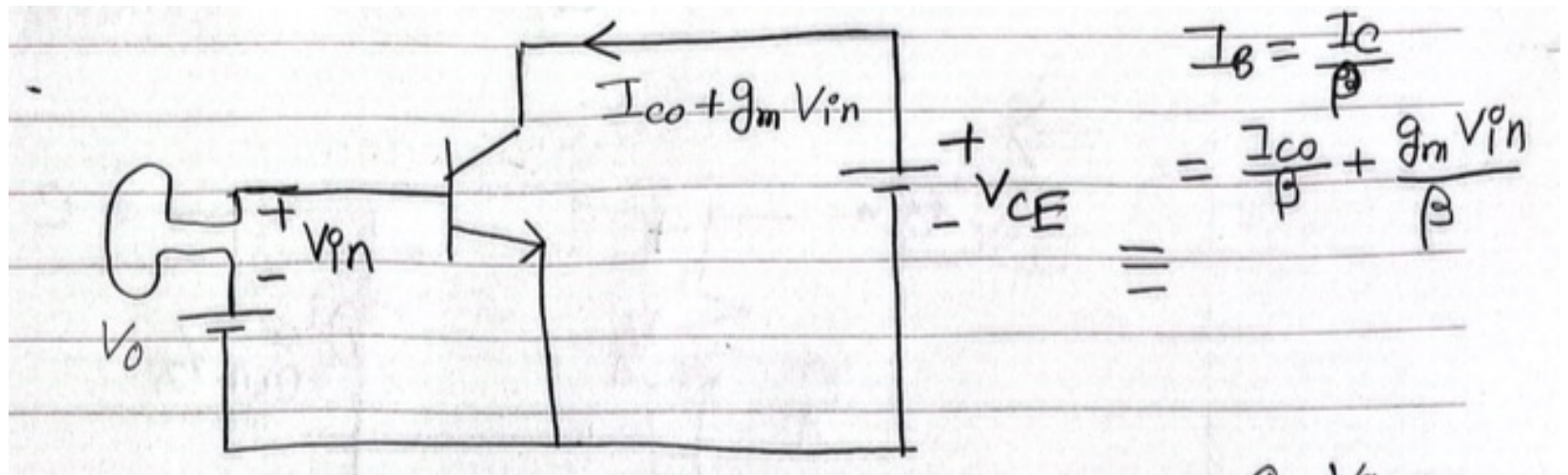
$$\textcircled{2} \quad I_c \approx I_{c0} + g_m V_m \sin \omega t$$



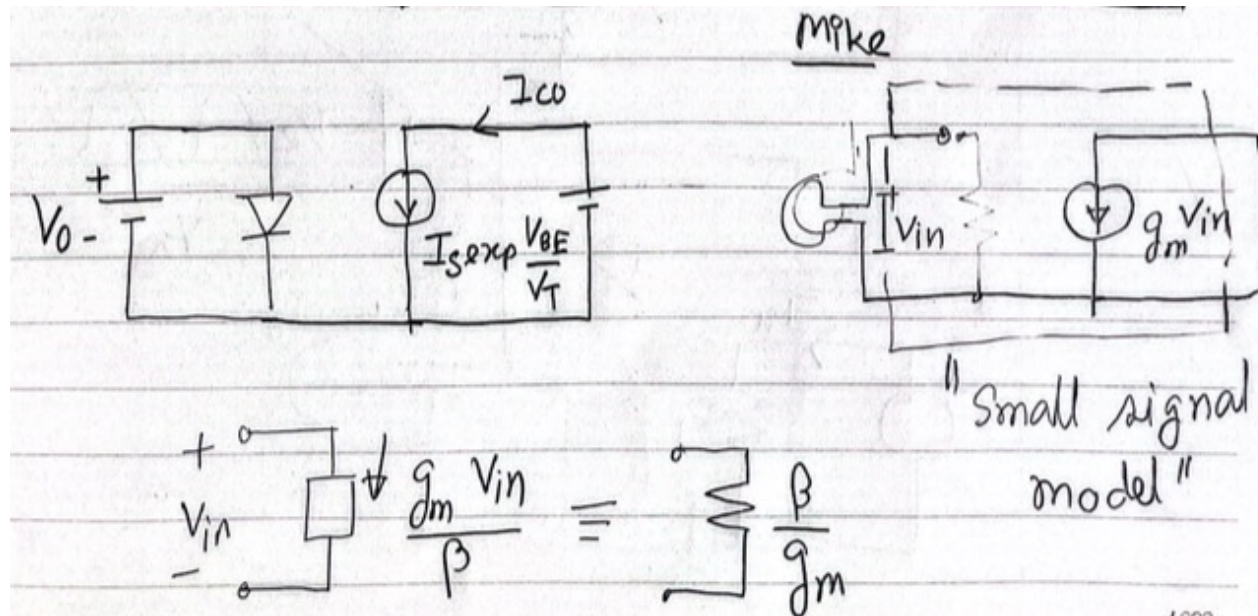
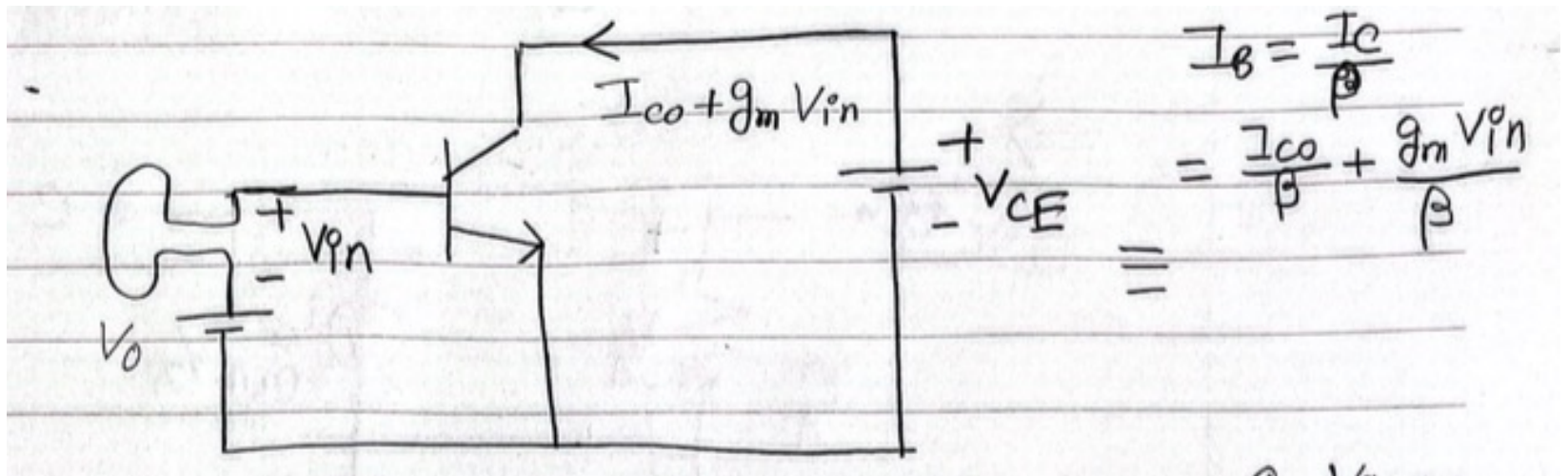
$$\text{derivative} = \frac{dI}{dV}$$

$$\therefore dI = dV g_m$$

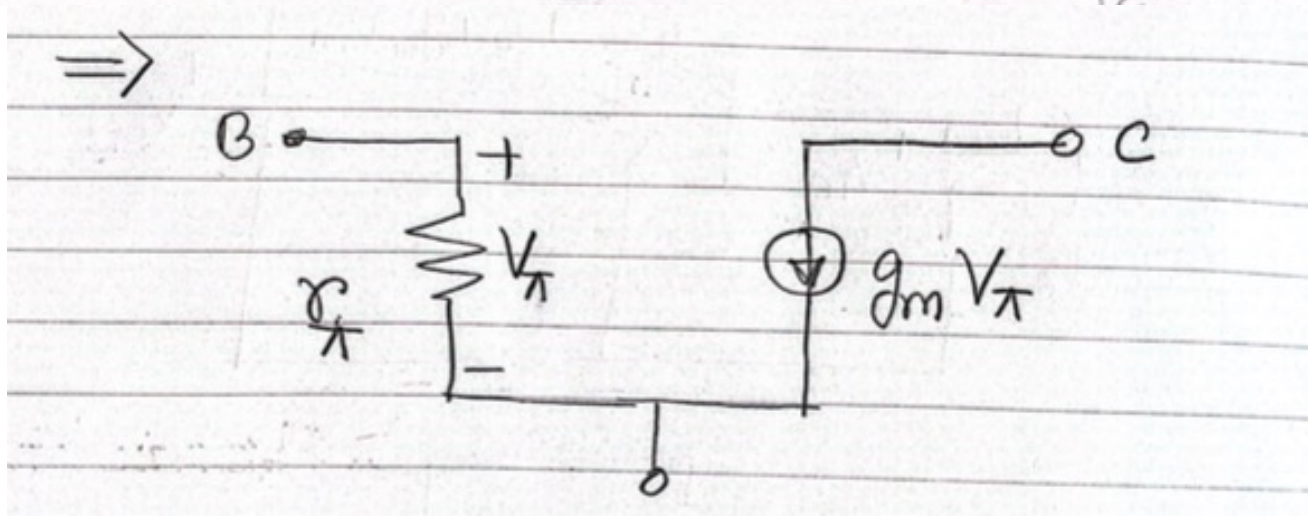
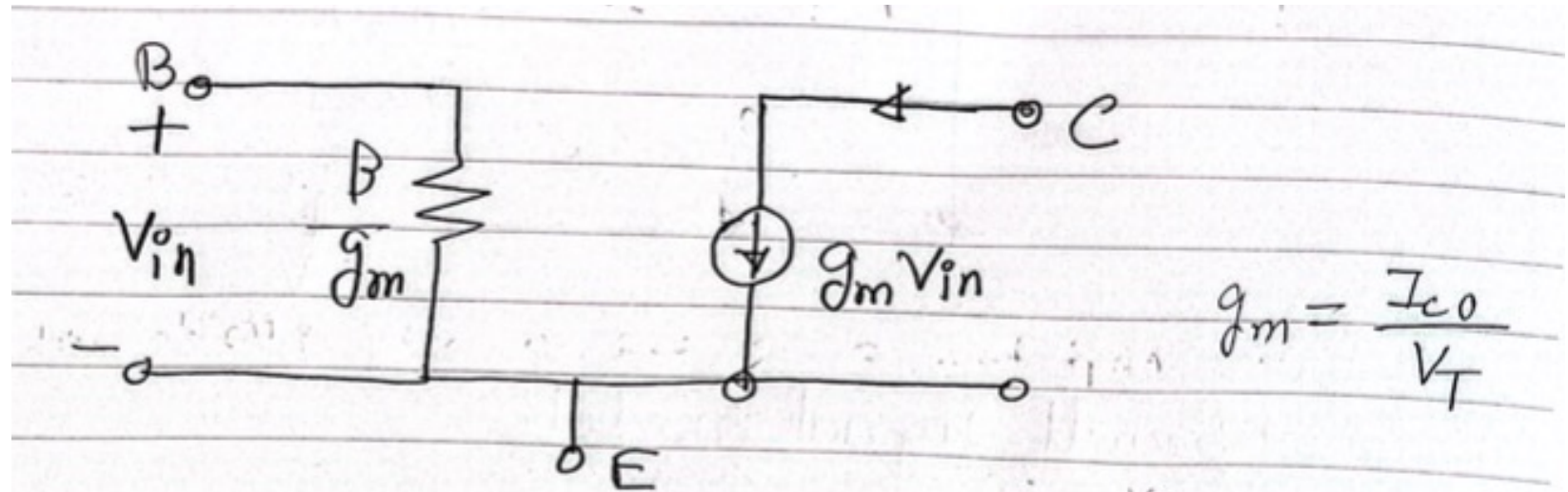
Bipolar Transistor Small Signal Model



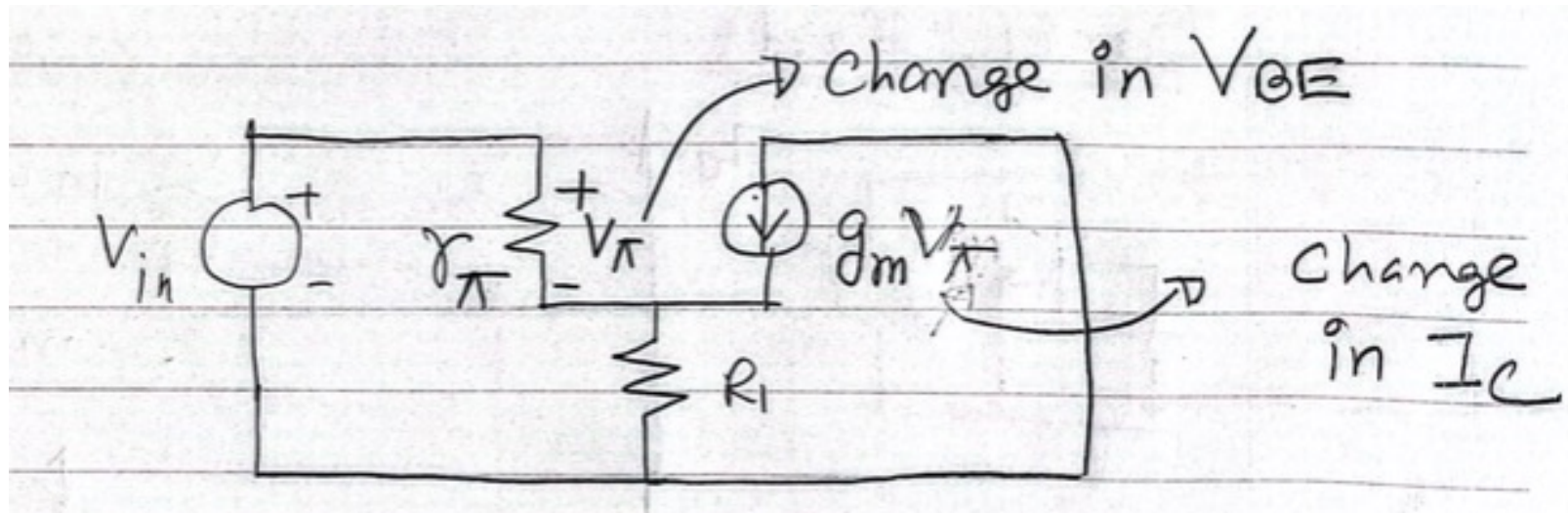
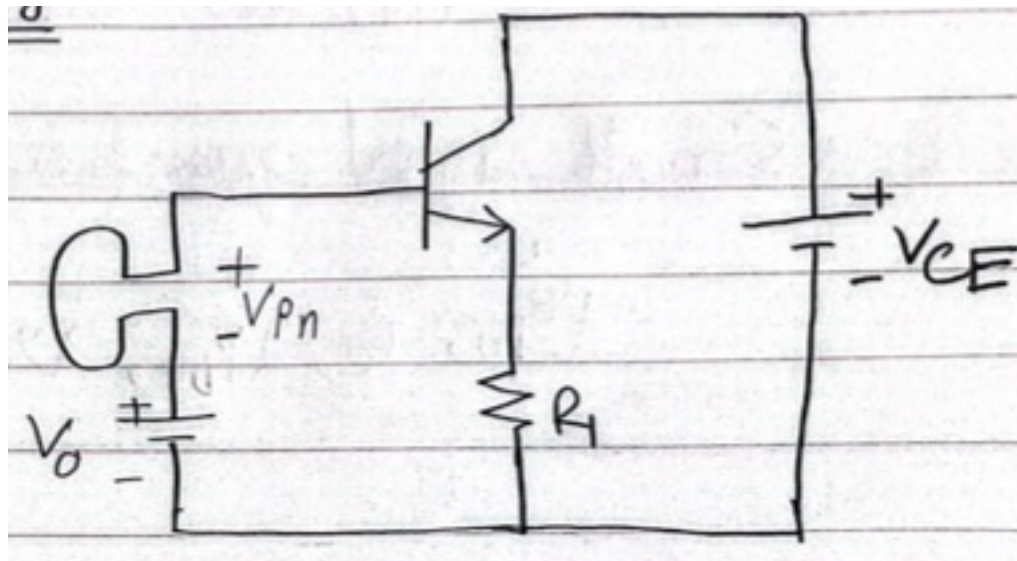
Bipolar Transistor Small Signal Model



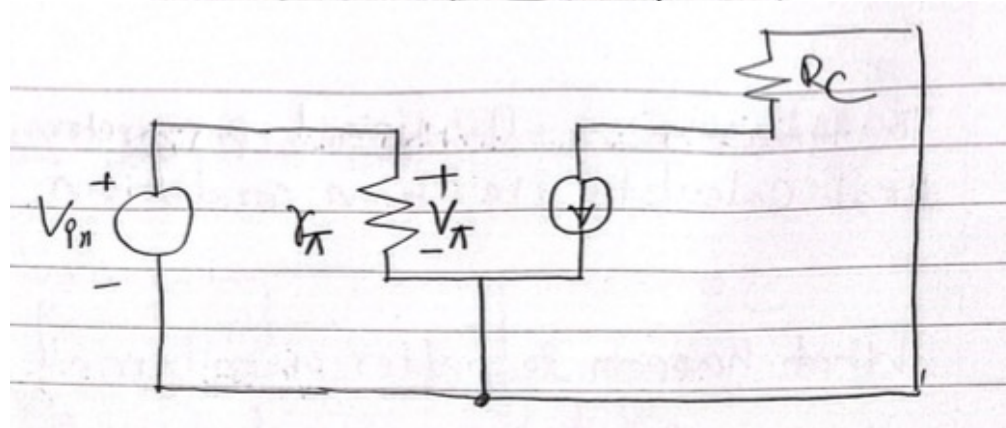
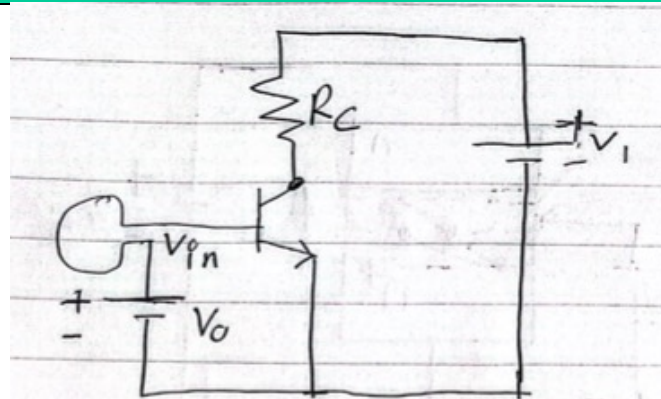
Complete Small Signal Model



Example: Small Signal Model



Example: Small Signal Model



Notation: Large-signal quantities: Uppercase.

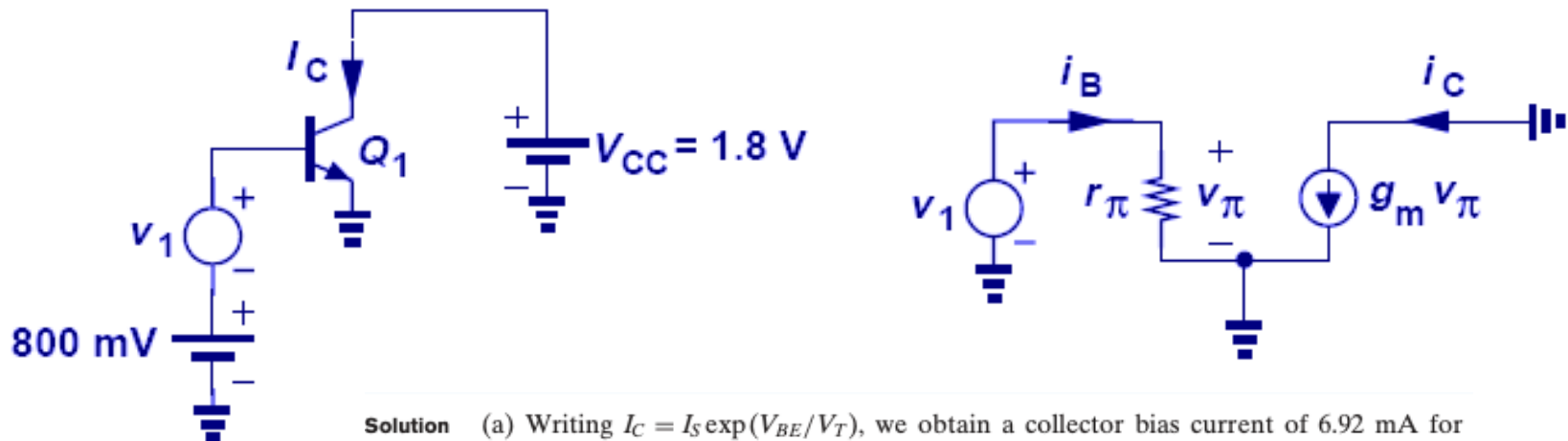
$V_{in}, I_c, V_{BE}, V_{CE}$

Small-signal quantities: lowercase.

$v_{in}, i_c, v_{be}, v_{ce}$

Small Signal Example I

Consider the circuit shown in figure below, where v_1 represents the signal generated by a microphone, $I_S = 3 \times 10^{-16} \text{ A}$, $\beta = 100$, and Q_1 operates in the active mode. If $v_1 = 0$, determine the small-signal parameters of Q_1 .



Solution (a) Writing $I_C = I_S \exp(V_{BE}/V_T)$, we obtain a collector bias current of 6.92 mA for $V_{BE} = 800 \text{ mV}$. Thus,

$$g_m = \frac{I_C}{V_T} \quad (4.54)$$

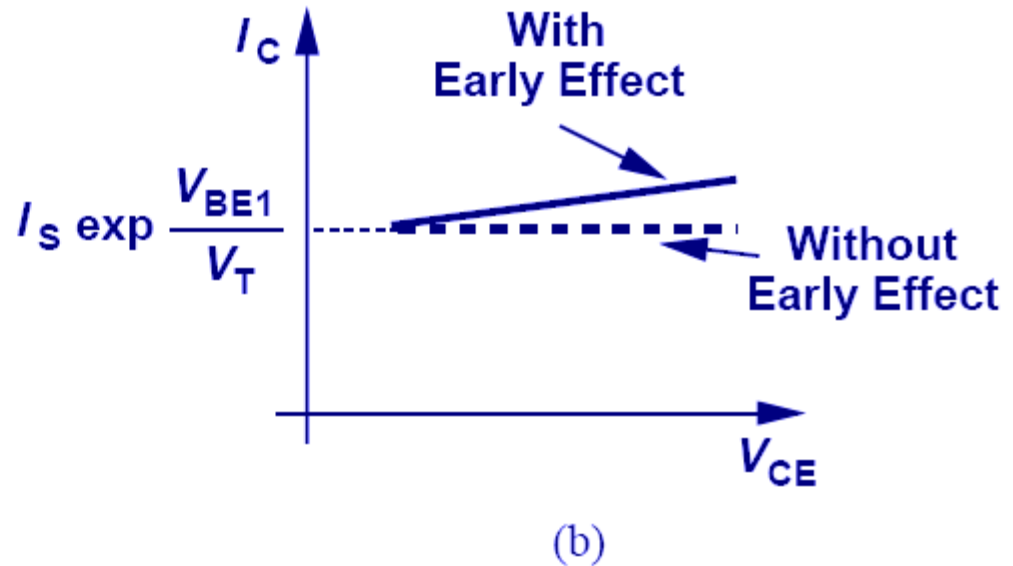
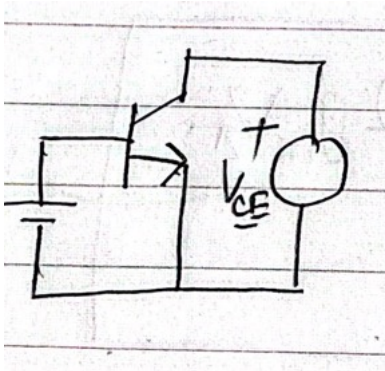
$$= \frac{1}{3.75 \Omega}, \quad (4.55)$$

and

$$r_\pi = \frac{\beta}{g_m} \quad (4.56)$$

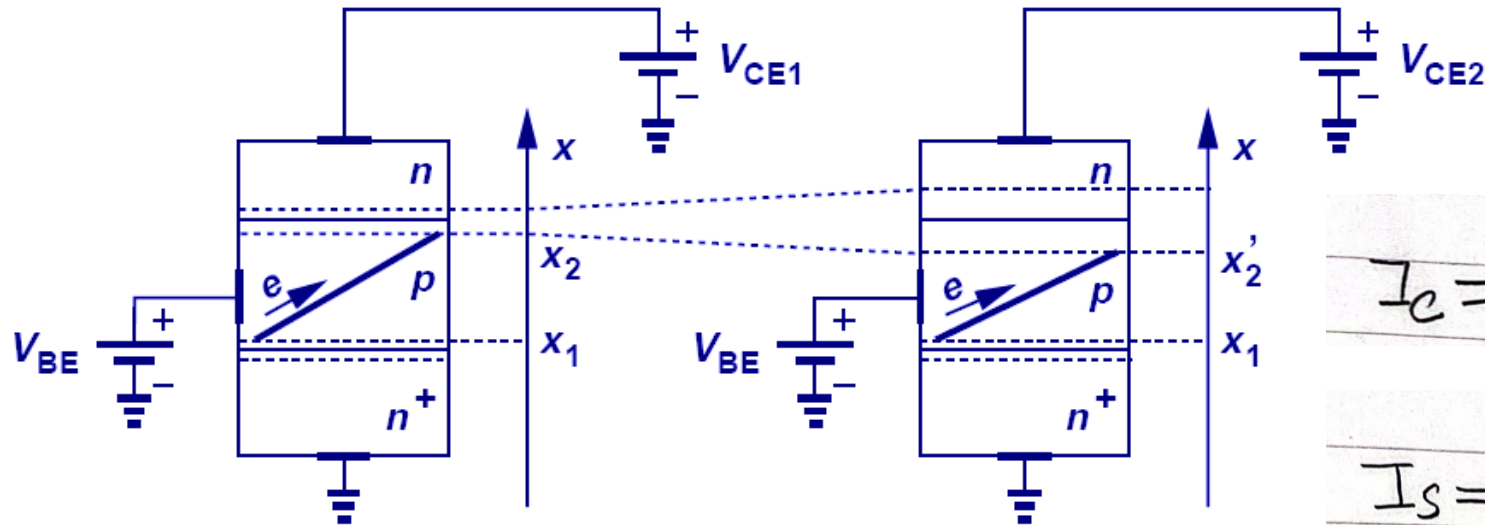
$$= 375 \Omega. \quad (4.57)$$

Early Effect



- **With Early effect, collector current becomes larger than usual and a function of V_{CE} .**

Early Effect

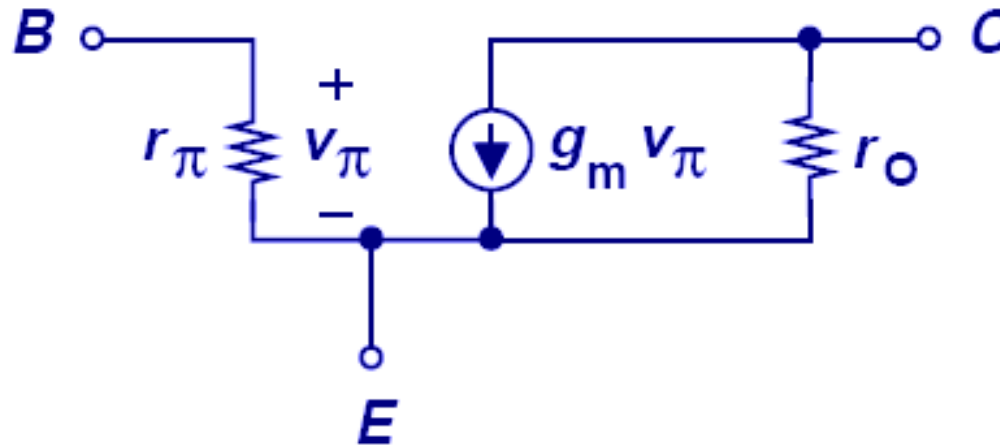


$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

$$I_S = \frac{A q n_i^2 \cdot D_n}{N_B W_B}$$

- The claim that collector current does not depend on V_{CE} is not accurate.
- As V_{CE} increases, the depletion region between base and collector increases. Therefore, the effective base width decreases, which leads to an increase in the collector current.

Final Small-Signal Model



$$g_m = \frac{I_C}{V_T} = \frac{1}{3.75\Omega}$$

$$r_\pi = \frac{\beta}{g_m} = 375\Omega$$

$$r_o = \frac{\Delta V_{CE}}{\Delta I_C} = \frac{V_A}{I_S \exp \frac{V_{BE}}{V_T}} \approx \frac{V_A}{I_C}$$

Example

A transistor is biased at a collector current of 1 mA. Determine the small-signal model if $\beta = 100$ and $V_A = 15$ V.

We have

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{1}{26 \, \Omega}, \end{aligned}$$

and

$$\begin{aligned} r_\pi &= \frac{\beta}{g_m} \\ &= 2600 \, \Omega. \end{aligned}$$

Also,

$$\begin{aligned} r_O &= \frac{V_A}{I_C} \\ &= 15 \, \text{k}\Omega. \end{aligned}$$